

Investigating the effect of interaction behavior on the ionic conductivity of Polyester/LiClO₄ blend systems

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ABSTRACT

We describe the interaction behavior between Polyester and Li⁺ that affects the ionic conductivity on binary electrolyte of lithium perchlorate (LiClO₄) blends with Poly(ethylene adipate), (PEA); Poly(1,4-butylene adipate), (PBA); Poly(1,6-hexamethylene adipate), (PHA); and Polycaprolactone (PCL), respectively. The individual four binary blend systems are similar and fully miscible, a different ionic conductivity exists in the binary Polyester electrolyte system corresponds to the result of the different interactions strength between lithium perchlorate (LiClO₄) and various Polyesters. The lithium ion is able to coordinate with more functional groups of PEA polymer chain than that in other three polyester chains. Both DSC and FTIR studies demonstrate that the ability of PEA to donate its electron to coordinate with Li⁺ is significantly stronger than that of PBA, PHA and PCL. Consequently, the maximum ionic conductivity ($1.06 \times 10^{-5} \text{ S cm}^{-1}$) at ambient temperature (30 °C) occurred at a composition of PEA/LiClO₄ (90/10).

It is further found that the ionic conductivity of Polyester electrolyte is not only dependent on the electron donation strength of carbonyl group, but also on the molecular structure of Polyester.

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1. Introduction

Solid polymer electrolytes (SPEs), which are complexes of solvent-free polymers and metal salts, are prepared by dissolving salts in high-molecular-weight polar polymer hosts. Polymer electrolytes have been studied extensively during the past two decades because of their potential applications, which include high energy density batteries and fuel cells [1–5]. The major motivation for this interest is the application in rechargeable and high energy density power sources. It is generally agreed that both of these qualities that achieved by the lithium-based electrochemical systems to configure the battery [6–8]. Ionic transport occurs in the polymer and very often is the result of a Li-salt dissolve and a coupling between the ions and segmental motions of the polymer chains. There are many methods have been applied to increase the ionic conductivity of the electrolyte, such as addition of a plasticizer to polymer matrix forming a gel polymer electrolyte [9–11]; addition of miner to polymer matrix forming a nanocomposites [12]; designing new polymeric matrices by synthetic processes or modifying existing polymer–salt complexes, and thus improve ionic conductivity [13–19].

It is an important challenge to develop practical methods for preparing the SPEs that have higher ionic conductivity and dimensional stability. In this regard, the preparation of polymeric electrolytes by choice an appropriate polymer structure is of interest. Polyester-based polymeric electrolytes are still among the most extensively studied polymer ionic conductors because their structures are beneficial for supporting fast ion transport. A strong electron donor within oxygen atom of carbonyl group in Polyester-based polymer electrolyte would tend to complex with lithium ion. The electron donation strength of carbonyl group would effect on the solubility of Li-salt and on the isolation of Li⁺ ion within the polymer matrix. However, the effect of electron donation strength of carbonyl group in Polyester on the ionic conductivity enhancement is not clearly understood.

There has been no previous study of the influence of the electron donation strength and its mechanism on the Polyester-based polymer electrolyte so far. A model system based on Adipate acidic Polyester doped with Lithium perchlorate (LiClO₄) examined in this work. The polyester of Polycaprolactone is added into this study to contrast the effect of polyester molecular structure on the ionic conductivity. The electron donation strength and isolation of Li⁺ ion within polymer electrolytes carbonyl group are detected by used differential scanning calorimetry (DSC), Fourier transform infrared spectroscopy (FTIR). Alternating current (AC) impedance

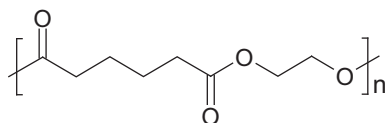
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investigates the electron donation strength and its related conductivity behavior of Polyester-based polymer electrolyte blend system. The rule to obtain is a higher ionic conductivity in Polyester-based polymer electrolyte blend system is presented.

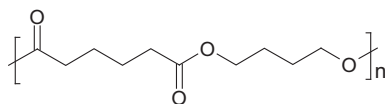
2. Experimental section

2.1. Materials

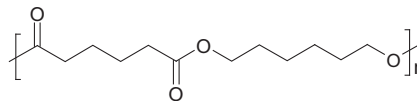
Poly(ethylene adipate) (PEA) with Mw of 10,000 and m.p. = 58–62 °C;



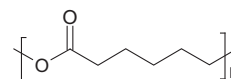
Poly(1,4-butylene adipate) (PBA) with Mw of 12,000 and m.p. = 56–60 °C;



Poly(1,6-hexamethylene adipate) (PHA) with Mw of 3800 and m.p. = 55–65 °C;



Polycaprolactone (PCL) with Mw of 65,000 and m.p. = 60 °C;



was purchased from Aldrich. All polymers without purification before used.

The Lithium perchlorate ($\text{LiClO}_4 \cdot 3\text{H}_2\text{O}$) salt, obtained from Acros, will be treated in 190 °C for 24 h in vacuum for eliminating crystal water and then stored in vacuum condition. Acetone, obtained from Aldrich, was refluxed at a room temperature under N_2 atmosphere before used.

2.2. Sample preparations

Polymer electrolytes of Polyester/ LiClO_4 in various blend compositions were prepared by solution casting. Desired amounts

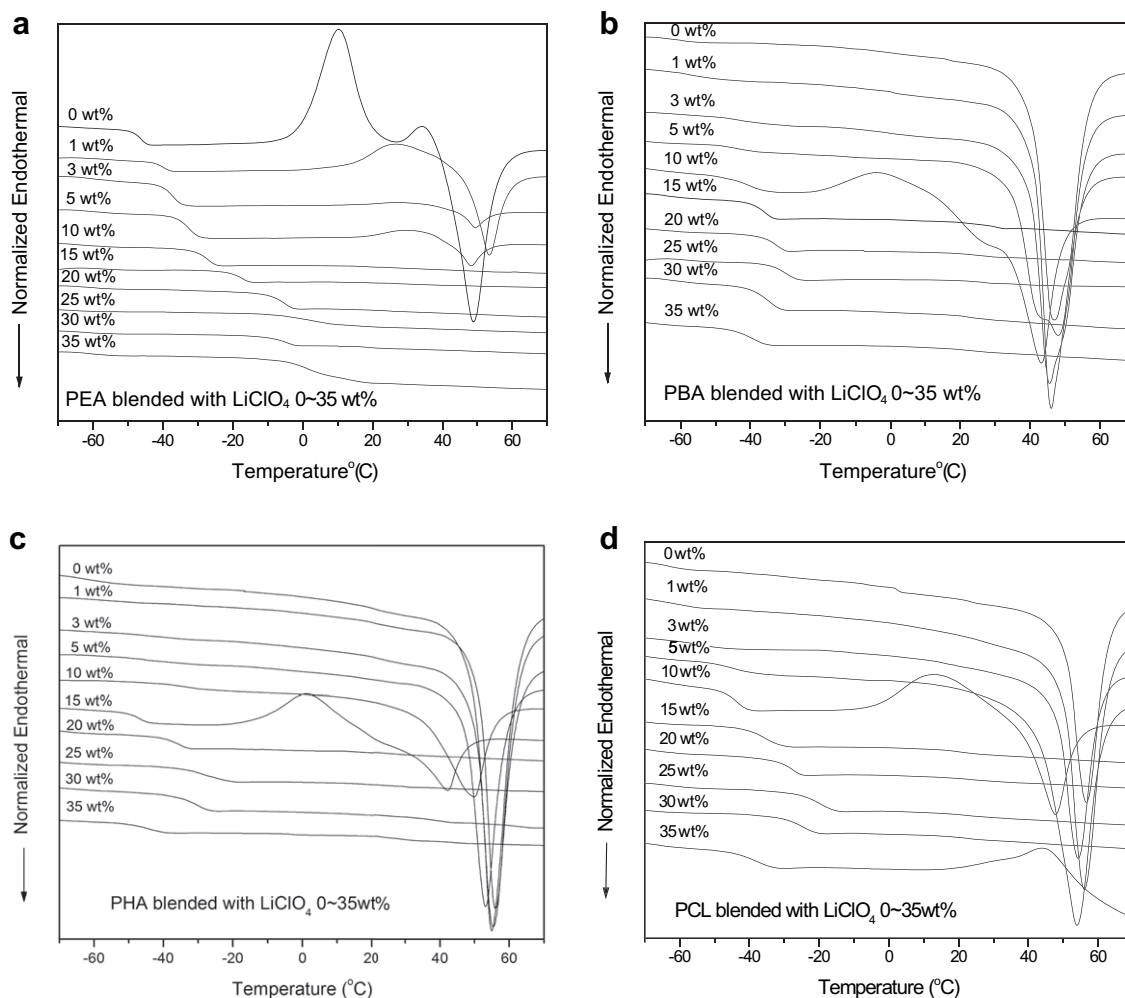


Fig. 1. (a): The second-run DSC thermograms of PEA blended with various LiClO_4 content ranging from 0 to 35 wt% contents. (b) The second-run DSC thermograms of PBA blended with various LiClO_4 content ranging from 0 to 35 wt% contents. (c) The second-run DSC thermograms of PHA blended with various LiClO_4 content ranging from 0 to 35 wt% contents. (d) The second-run DSC thermograms PCL blended with various LiClO_4 content ranging from 0 to 35 wt% contents.

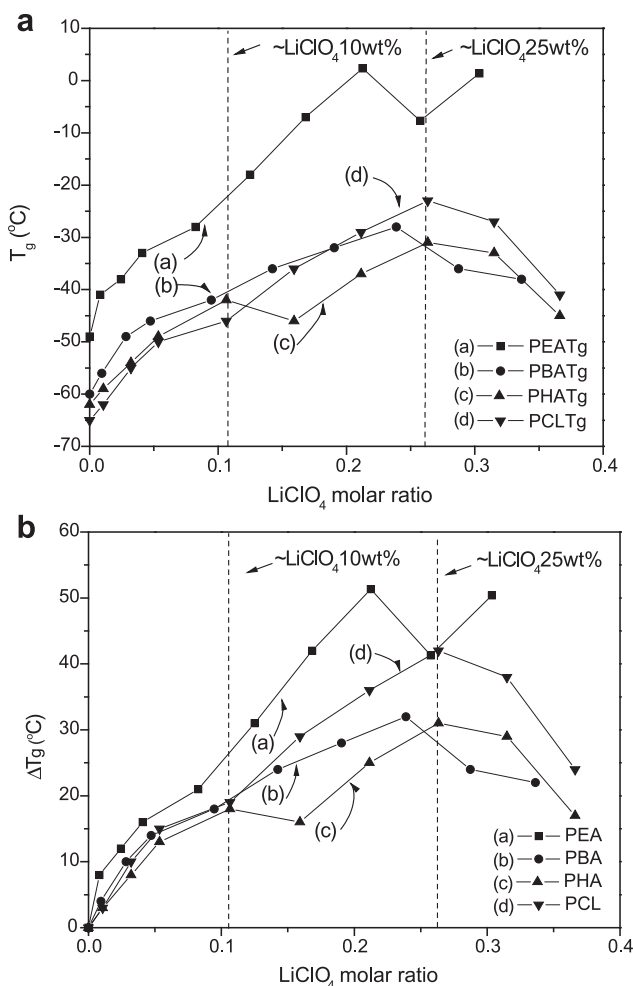


Fig. 2. (a): Summarization of glass transition temperature of (a) PEA/ LiClO_4 ; (b) PBA/ LiClO_4 ; (c) PHA/ LiClO_4 and (d) PCL/ LiClO_4 blends on the molar basis. (b) Deviation of glass transition temperature of (a) PEA/ LiClO_4 ; (b) PBA/ LiClO_4 ; (c) PHA/ LiClO_4 and (d) PCL/ LiClO_4 blends on the molar basis.

of PEA, PBA, PHA, PCL, and LiClO_4 salt were dissolved in dry acetonitrile and stirred continuously for 24 h at 60 °C. The solution was cast onto a Teflon dish and maintained at 50 °C for an additional 24 h to remove the solvent, and then the dish was further dried under vacuum at 80 °C for 2 days. To prevent its contact with the air and moisture, the polymer electrolyte was transferred to a glovebox under a nitrogen atmosphere.

2.3. Differential scanning calorimetry (DSC)

Thermal analyses were performed using a DSC instrument (DuPont TA 2010). The sample was heated from −100–150 °C under dry nitrogen. The glass transition temperature (T_g) was obtained as the inflection point of the heat capacity jump recorded at a scan rate of 20 °C min^{-1} .

2.4. Conductivity measurements

Ionic conductivity measurements with alternation current were conducted on an AUTOLAB designed by Eco Chemie within the frequency range from 10 MHz to 10 Hz. The electrolyte film was sandwiched between stainless steel blocking electrodes (diameter: 1 cm). The specimen thickness varied from 0.8 to 1.2 mm; the

impedance response was gauged over the range from 20 to 120 °C. The bulk resistance was derived from the Cole–Cole plot of the complex impedance data of the blend where the imaginary impedance is zero [20–22]. Conductivity was calculated from the bulk resistance according to the equation:

$$\sigma = \frac{l}{RA}$$

where σ is the conductivity, l is the thickness of the electrolyte film, A is the section area of the stainless steel electrode, and R is the bulk resistance.

3. Results and discussion

3.1. DSC studies

The DSC analysis is one of the most convenient methods for determining the miscibility in polymer blends. Fig. 1a–d present the conventional second-run DSC thermograms of PEA, PBA, PHA and PCL blended with various LiClO_4 content ranging from 0 to 35 wt% contents, respectively. T_g of the pure polymers used in this study, PEA, PBA, PHA and PCL, are ca. −49, −60, −62 and −65 °C, respectively. The observed T_g is changed and identified with the increase of LiClO_4 content in all blends. A changed T_g strongly suggests that these blends are miscible and exist an interaction within the homogeneous amorphous phase in this study. It indicates that these interactions can be grouped into Polyester/ LiClO_4 complex in the four Polyester/ LiClO_4 blends.

Fig. 2a and b display the summarization of glass transition temperatures and its deviation of (a) PEA/ LiClO_4 ; (b) PBA/ LiClO_4 ; (c) PHA/ LiClO_4 and (d) PCL/ LiClO_4 blends on the weight basis, respectively. The T_g elevation patterns of three adipate acid Polyesters look similar when the LiClO_4 lowers than 20 wt%; notably, the PCL/ LiClO_4 shows a slight different from the three others. In the four blend systems, the T_g s increased with increasing the LiClO_4 content in initial stage, but its T_g tends to decrease and separate into two T_g s when the concentrations of LiClO_4 is more than 25 wt% for all four blend systems, as seen in Fig. 1a–d. It suggested that the T_g increased initially cause of that the LiClO_4 salt reduces the mobility of polyester polymer chain; whereas the decreased in T_g when LiClO_4 contents is more than 25 wt% due to there is repulsion between the free anions or the dilution effect of salt aggregation [23,24]. There are two phases would be formatted at high LiClO_4 contents region, concluded from the double T_g behavior.

The order of T_g increment of the three adipate polyester blends is PEA/ LiClO_4 > PBA/ LiClO_4 > PHA/ LiClO_4 for the whole blend range. The PCL/ LiClO_4 blend is between PBA/ LiClO_4 and PHA/ LiClO_4 when the LiClO_4 is less 10 wt% content; and between PEA/ LiClO_4 and PBA/ LiClO_4 when the LiClO_4 is within 10–20 wt%, as shown in Fig. 2b. The elevation maximum in PEA/ LiClO_4 , PBA/ LiClO_4 , PHA/ LiClO_4 , and PCL/ LiClO_4 takes place at Li/O mole ratio of 0.21, 0.23, 0.28, and 0.27 with T_g deviation of 48, 30, 30 and 40, respectively. It is understandable because PEA interacts stronger with lithium ions than the other three blends. Moreover, the distance of ester group in PEA is the shortest than that in other three polyester polymers. It has been suggested that the lithium ion is able to coordinate with more functional groups of the polymer chain in PEA than that in other three polyester chains. As a result, a lower molar ratio of LiClO_4 salt can achieve maximum T_g deviation in the PEA/ LiClO_4 system. The PCL/ LiClO_4 shows intermediate level interaction strength and different behavior within the four blend systems, as shown in Fig. 2.

Clarifying further the interaction phenomenon of Li-salt in different Polyester, the endothermic peak behavior of blends are studied as followings: the interaction format the transient

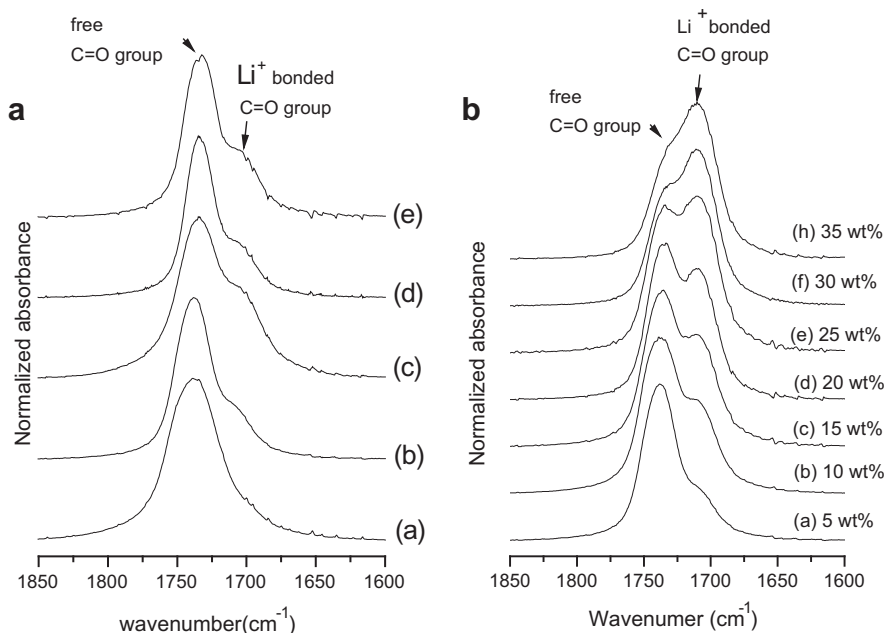


Fig. 3. (A): Infrared spectra of (a) pure PEA; and a series of Polyester/LiClO₄ binary blends in specific amount of LiClO₄ (5 wt%) recorded at 80 °C. (b) PEA, (c) PBA, (d) PHA and (e) PCL. (B) Infrared spectra of a series of PEA/LiClO₄ binary blends contains (a) 5 wt%, (b) 10 wt%, (c) 15 wt%, (d) 20 wt%, (e) 25 wt%, (f) 30 wt%, and (h) 35 wt% recorded at 80 °C.

crosslinks between the salt and the polyester phase result in lower crystallinity. Some differences on the endothermic peak behavior can be observed, as shown in Fig. 1a–d. The endothermic peak closes to ~49 °C observed in Fig. 1a, corresponding to the melting of pure PEA crystalline phase. When the LiClO₄ is added, the endothermic peak decreased progressively and shifts toward lower temperature. The change of the PEA thermal properties is attributed to the incorporation of the LiClO₄ salt, which interferes with the PEA chain arrangement and its crystallinity. As observed from Fig. 1a–d, it is found that the four T_m s decreasing pattern look similar, the T_m disappeared completely at LiClO₄ content of around 10 wt%, 15 wt%, 20 wt% and 15 wt% for PEA/LiClO₄, PBA/LiClO₄, PHA/LiClO₄ and PCL/LiClO₄ blend, respectively. In other words, the crystallinity in PEA/LiClO₄ blend disappeared completely with the minimum LiClO₄ amount in the four Polyester/LiClO₄ systems. It is suggested that the lithium ion coordinates with the PEA to retards the crystallinity of PEA is more serious than the other three polyesters.

In the other hand, as can be seen, the order of heat enthalpy is PHA/LiClO₄ > PCL/LiClO₄ > PBA/LiClO₄ > PEA/LiClO₄ when the LiClO₄ content is 10 wt%, as shown in Fig. 1a–d. The recrystallization of PHA/LiClO₄ would be occurred till the LiClO₄ content is higher than 20 wt%; higher than other three Polyester/LiClO₄ systems. It means that the LiClO₄ interferes with the PHA chain arrangement and its crystallinity is not as serious as in other three blend systems. Seemingly, the lithium ion interacts with PHA chain is weaker than the other three systems. As a result, the crystallinity of PHA has to be eliminated by the highest LiClO₄ salt content within the Polyester/LiClO₄ blend system.

3.2. FT-IR spectroscopy

Since the carbonyl group (C=O) is a strong electron donor within the Polyester-based polymer electrolyte, the lithium ion tends to complex with the oxygen atom of the carbonyl group. Infrared spectrum is a powerful tool to monitor such ionic interaction and morphology of polymer blend. Fig. 3a shows infrared

spectra recorded at 80 °C representing the carbonyl stretching of a series of Polyester blended with LiClO₄ 5 wt% content. Fig. 3b shows infrared spectra recorded at 80 °C representing the carbonyl stretching of PEA/LiClO₄ binary blends with 5, 10, 15, 20, 25, 30 and 35 wt% LiClO₄ contents. According to previous publications [24–26], the peak at ~1730 cm⁻¹ in Fig. 3a and b, represents the free C=O group of the Polyester. When the LiClO₄ salt is added, as shown in Fig. 3a and b, a weak shoulder at 1700 cm⁻¹ appears which reflects the interaction between the Li⁺ ion and carbonyl

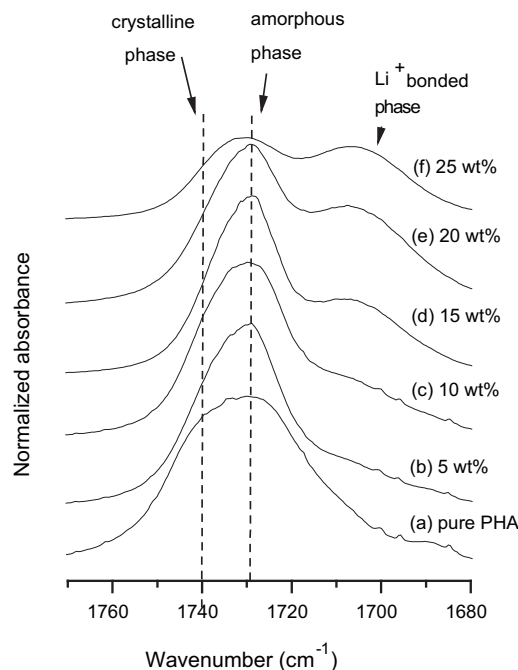


Fig. 4. The C=O-stretching of PHA/LiClO₄ binary blends with various amount of LiClO₄ at 30 °C cooled from 80 °C. (a) 5 wt%, (b) 10 wt%, (c) 15 wt%, (d) 30 wt% and (e) 35 wt%.

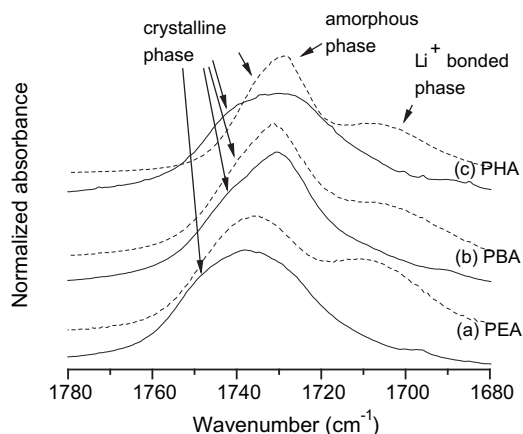


Fig. 5. Infrared spectra recorded at 30 °C cooled from 80 °C, displaying the bands representing the C=O-stretching of (a) PEA/LiClO₄ (b) PBA/LiClO₄ (c) PHA/LiClO₄ blend system with LiClO₄ 0 wt% (solid line) and 15 wt% (dot line).

group of Polyester. This shoulder peak grown upon the addition of LiClO₄, as shown in Fig. 3b, indicating there is the interaction between Li⁺ ion and PEA grow with the addition of LiClO₄.

Fig. 4 shows infrared spectra recorded at 30 °C cooled from 80 °C, displaying the bands representing the C=O-stretching of PHA/LiClO₄ binary blends with various LiClO₄ contents. There are two C=O-stretching bands appear for the pure PHA at ~1740 and ~1730 cm⁻¹; these bands correspond to the absorptions of the crystalline and amorphous conformations, respectively. It is found that the relative peak area of the crystalline conformation of PHA (~1740 cm⁻¹) decreases upon increasing the LiClO₄ content in this PHA/LiClO₄ blend system at room temperature. The crystalline conformation of PHA is partially destroyed when the LiClO₄ content is increased in this electrolyte blends system. As a result, we confirm that the interaction between PHA and LiClO₄ tends to decrease the fraction of the crystalline phase of PHA.

Fig. 5 shows infrared spectra recorded at 30 °C cooled from 80 °C, displaying the bands representing the C=O-stretching of (a) PEA/LiClO₄ (b) PBA/LiClO₄ (c) PHA/LiClO₄ blend system with LiClO₄ 0 wt% and 15 wt% content, respectively. The crystalline phase of PEA nearly disappears when the LiClO₄ 15 wt% is added, as shown in

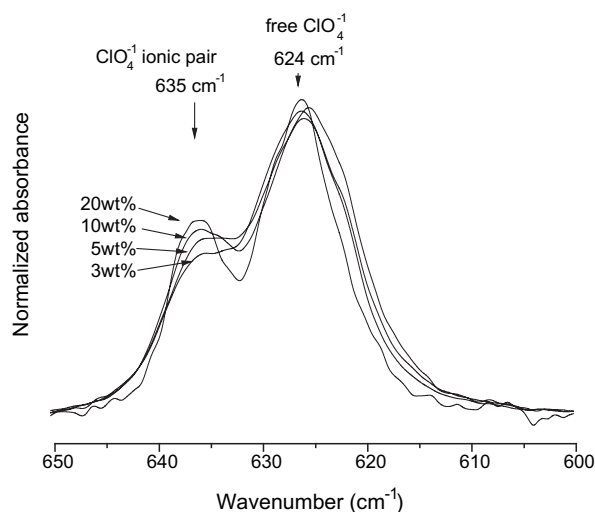


Fig. 6. Infrared spectra in the (ClO₄⁻) spectral ranging from 650 to 600 cm⁻¹ containing various LiClO₄ concentrations for PHA/LiClO₄ binary blends at 80 °C where a larger adsorption band at 624 cm⁻¹ and a smaller one at 635 cm⁻¹.

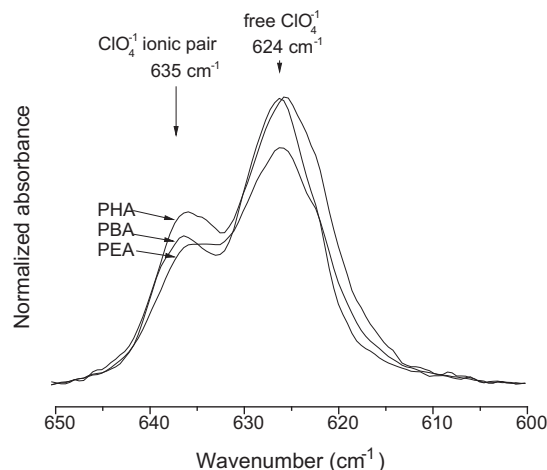


Fig. 7. Infrared spectra in the (ClO₄⁻) spectral ranging from 650 to 600 cm⁻¹ where PEA, PBA, and PHA blended with LiClO₄ 10 wt% at 80 °C.

Fig. 5a. By comparing these spectra with those in Fig. 5, the relative strength of crystalline PHA phase in this Polyester/LiClO₄ blend system the highest one at the LiClO₄ 15 wt% content. On the other hand, the relative strength of Li⁺ bonded C=O of PEA is higher than the other two, the order is PEA > PBA > PHA at the LiClO₄ 15 wt% content, as shown in Fig. 5. It is clear to identify that the addition of LiClO₄ tends to interact with PEA to destroy the PEA crystalline more seriously than the others; the lithium perchlorate prefers to coordinate with PEA rather than with PBA and PHA. This finding suggests that the LiClO₄ content in the PEA/LiClO₄ blend system interacts with or complexes to all available PEA essentially, which,

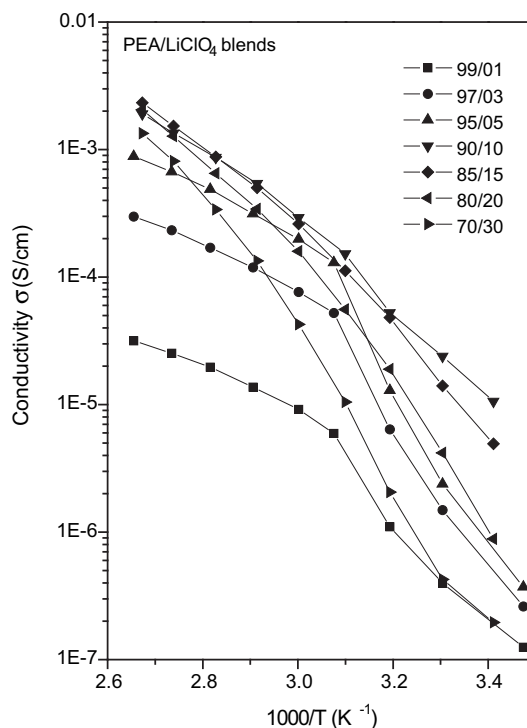


Fig. 8. Arrhenius conductivity plots for PEA/LiClO₄ electrolyte composites containing various LiClO₄ concentrations. PEA/LiClO₄ = (■) 1 wt%; (●) 3 wt%; (▲) 5 wt%; (▼) 10 wt%; (◆) 15 wt%; (◀) 20 wt%; (▶) 30 wt%.

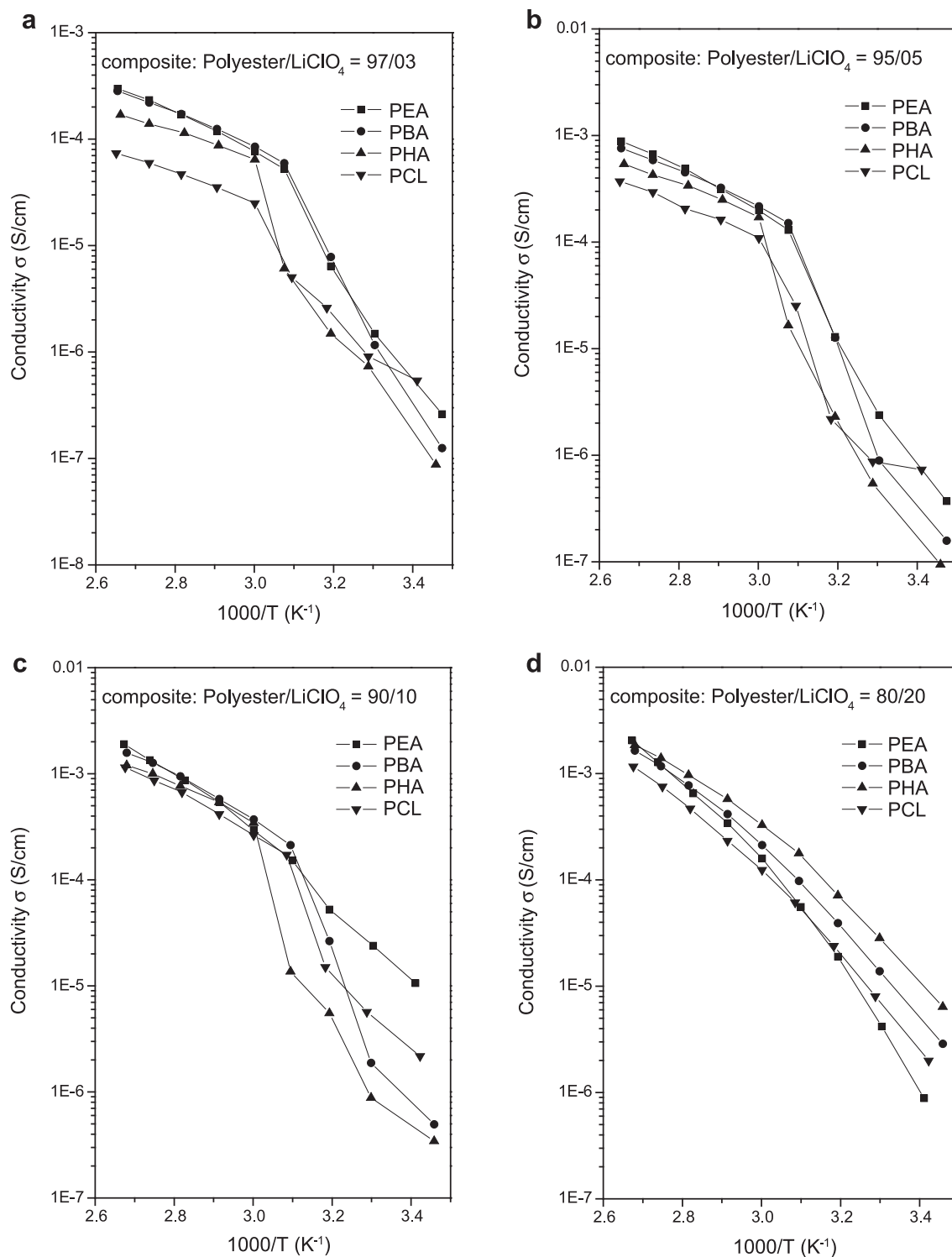


Fig. 9. (a): Arrhenius conductivity plots for Polyester/LiClO₄ (3 wt%) electrolyte composite: (■) PEA; (●) PBA; (▲) PHA; (▼) PCL. (b) Arrhenius conductivity plots for Polyester/LiClO₄ (5 wt%) electrolyte composite: (■) PEA; (●) PBA; (▲) PHA; (▼) PCL. (c) Arrhenius conductivity plots for Polyester/LiClO₄ (10 wt%) electrolyte composite: (■) PEA; (●) PBA; (▲) PHA; (▼) PCL. (d) Arrhenius conductivity plots for Polyester/LiClO₄ (20 wt%) electrolyte composite: (■) PEA; (●) PBA; (▲) PHA; (▼) PCL.

thus, retards its crystal formation. The interaction behavior of LiClO₄ with Polyester is consistent to the DSC studies.

We discuss the dissolution of lithium perchlorate salts. The characteristic ν (ClO₄⁻) mode of the lithium perchlorate (LiClO₄) is particularly sensitive in changing the local anionic environment [25–28]. According to prior literature [25,28], the 624 cm⁻¹ has

been assigned to the free anion which does not interact directly with lithium ion. Component at 635 cm⁻¹ is attributed to contact ion pair. Fig. 6 shows the infrared spectra in the (ClO₄⁻) spectral ranging from 650 to 600 cm⁻¹ containing various LiClO₄ concentrations for PHA/LiClO₄ binary blends at 80 °C where a larger adsorption band at 624 cm⁻¹ and a smaller one at 635 cm⁻¹. As

shown in Fig. 6, the relative strength of 635 cm^{-1} absorption increases gradually with increasing LiClO_4 concentration. It means that the fraction of ion-pair anion is increased by the increasing LiClO_4 content. The fraction of ion-pair anions increased more rapidly when the amount of LiClO_4 approaches at about 20 wt%. It can be attributed to higher LiClO_4 salt concentration in PHA reduces the effects of dissolving.

Fig. 7 shows the infrared spectra in the (ClO_4^-) spectral ranging from 650 to 600 cm^{-1} where PEA, PBA, and PHA blended with LiClO_4 10 wt% at 80°C , respectively. It is clear to observe that the relative strength of 635 cm^{-1} band in PHA is higher than that in PEA and PBA. It means that the highest fraction of ion-pair anions appears on the PHA, the second is PBA, and the last one is PEA. In the other word, the highest fraction of free ionic anions would appear in the PEA polymer matrix at the same LiClO_4 contents.

The PEA shows a highest ability to dissolve the lithium salts, the phenomenon is consistent to the result of T_g and $\text{C}=\text{O}$ FTIR spectra. The fraction of free ion anions in PEA polymer matrix is higher than other because essentially the $\text{C}=\text{O}$ group of electron donate to Li^+ in shorter PEA repeated unit is stronger than the longer repeated one (PBA, and PHA).

From the both DSC and FTIR studies suggest and demonstrate that the ability of PEA to donate its electron to Li^+ is significantly stronger than that of PBA and PHA. We will further discuss the effects of electron donation on the ionic conductivity.

3.3. Ionic conductivity

The ionic conductivity σ of an electrolyte is defined as the product of concentration of ionic charge carriers and their mobility

$$\sigma = \sum n_i z_i \mu_i$$

where n_i , z_i , and μ_i refer to the number of charge carriers, the ionic charge, and the ionic mobility, respectively. According to the aforementioned equation, the ionic conductivity depends on the number of charge carriers (n_i) in the system and the mobility (μ_i) of the various species.

Fig. 8 presents Arrhenius plots that illustrates the temperature dependency of the ionic conductivity for PEA/ LiClO_4 electrolyte composites containing various LiClO_4 concentrations. The conductivity increases with the increase of the LiClO_4 contents and attains a maximum value when the LiClO_4 is round 10–15 wt%. Subsequently, the conductivity decreases drastically with further increases of the further increases of the LiClO_4 concentrations. The addition of LiClO_4 in the binary blends tends to retard or destroy the crystalline phase and results in higher ionic conductivity as a result of the strong interaction between PEA and LiClO_4 . In addition, the crystalline phase can be reduced or inhibited and higher conductivity occurs with raising the temperature, as is expected. If we consider, for example, a composition of PEA/ LiClO_4 (3 wt%), we observe that its ionic conductivity increases drastically when the temperature is raised from 40 to 60°C . However, further addition of the LiClO_4 salt in this Polyester-base in the system may aggregate and tightly hold the Li^+ ion that would reduce ionic mobility, thereby reducing the observed conductivity. As a result, the maximum ionic conductivity occurs at the composition at which the LiClO_4 begins to aggregate.

Fig. 9 presents the Arrhenius plots that illustrates the temperature dependence on the ionic conductivity of various Polyester/ LiClO_4 compositions. It is found that the PEA/ LiClO_4 blend exhibits the highest conductivity when the LiClO_4 is up to 10 wt%, and higher than that of PHA/ LiClO_4 blend 0.5–2 orders in whole temperature range, as shown in Fig. 9a, b and c. Notably, at 30°C (room temperature), the highest conductivity ($1.06 \times 10^{-5}\text{ S cm}^{-1}$)

is achieved when PEA blended with LiClO_4 10 wt%, which is 30 times higher than the PHA blended with the same LiClO_4 content, as shown in Fig. 9c.

The PEA/ LiClO_4 has the lowest conductivity in room temperature when the LiClO_4 is 20 wt% content, as shown in Fig. 9d. The LiClO_4 salt in PEA has been aggregated, however, the PHA is not occurred on the same LiClO_4 content in room temperature due to the LiClO_4 has been not aggregated. Consequently, the conductivity of PHA is expected to have the highest at LiClO_4 20 wt% content.

In additional, the conductivity behavior of PCL/ LiClO_4 , the ester density is equal to PHA, is rather different from that of other three Polyesters due to different polymer structure and electron donation behavior of ester group in PCL. Thus, to achieving the highest possible ionic optimum environment for ionic transport and the highest possible ionic conductivity have to consider the balanced interaction among the Li^+ , LiClO_4 salt, ether groups, anions, structure of polymer matrix etc... It would result in an optimum environment for ionic transport and achieve the highest possible ionic conductivity.

4. Conclusions

We have investigated the effect of interaction mechanism on ionic conductivity of Polyester/ LiClO_4 binary blend-based electrolyte systems by the use of DSC, FTIR spectroscopy, and AC impedance measurements. Although each of the four individual binary pairs is fully miscible, a similar interaction behavior exists in four Polyester blend systems. Li^+ ion more preferably coordinates with the carbonyl group atom of PEA rather than with that of the other three polyesters, it is because that the $\text{C}=\text{O}$ group of electron donate essentially to Li^+ in shorter PEA repeated unit is stronger than the longer repeated one. When LiClO_4 is added to the Polyester/ LiClO_4 binary blends, the carbonyl group of PEA tends to interact with Li^+ more strongly than that of other three Polyester polymers, which causes the PEA crystallization is retarded and inhibited more seriously than other three. This factor is responsible for the observed highest ionic conductivity in PEA/ LiClO_4 blend due to PEA/ LiClO_4 blend own the highest carrier in the same LiClO_4 contents. The further addition of the LiClO_4 salt in this Polyester-base in the system may aggregate and tightly hold the Li^+ ion that would reduce ionic mobility, thereby reducing the observed conductivity. Therefore, a maximum ionic conductivity ($1.06 \times 10^{-5}\text{ S cm}^{-1}$) of Polyester/ LiClO_4 blend at room temperature (30°C) occurs at the PEA/ LiClO_4 composition by LiClO_4 10 wt%.

Furthermore, we found that not only electrolyte donation strength, but the molecular structure also effect on the ionic conductivity slightly. It has to find the ionic optimum environment for ionic transport to consider the balanced interaction among the Li^+ , LiClO_4 salt, ether groups, anions, structure of polymer matrix etc...

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